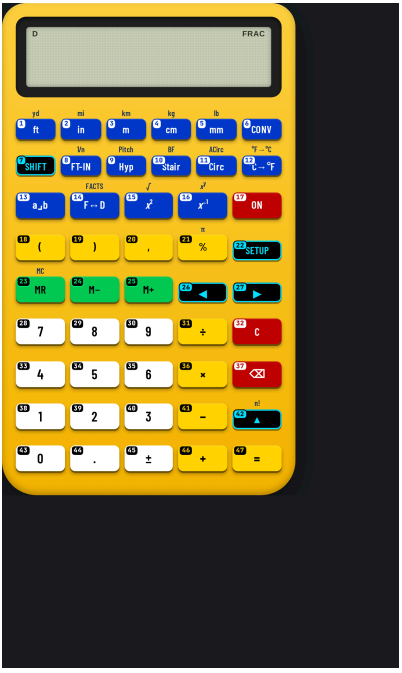
U-45 CALCULATORS

User Manual

Calculator #28 Fraction & Units One-Line (paper white)

Scientific – classic one-line



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#28 — Fraction & Units One-Line (paper white)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example  . Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

one-line algebraic: exact fractions with a␣b␣c and measurement units (type, then =).

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers
12. Appendix B — Function finder (A–Z)

13. Appendix C — Glossary

14. Appendix D — Model notes & limits

15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#28** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#28, #29, #30, #33, #34, #35, #38, #39, #40, #43, #44, #45
op	Operators & execute	#18, #19, #20, #21, #31, #36, #41, #46, #47
fn	Functions	#1, #2, #3, #4, #5, #6, #8, #9, #10, #11, #12, #13, #14, #15, #16
mem	Memory, variables & registers	#23, #24, #25
clr	Clear, delete & power	#17, #32, #37

mod	**Shift, modes, menus & navigation**	**#7, #22, #26, #27, #42**

2 · Features

44 features: 30 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Exact fraction arithmetic (no rounding): 1␣3 + 1␣6 = 1␣2**
2. LIVE **Mixed numbers a␣b␣c entry**
3. LIVE **Result cycle: improper fraction → mixed number → decimal (F ↔ D)**
4. LIVE **Automatic lowest terms**
5. LIVE **A decimal point switches that calculation to decimals automatically**
6. LIVE **Fraction-facts screen: mixed, decimal, numerator, denominator, percent, reciprocal**
7. LIVE **Length units on keys: ft, in, m, cm, mm (yd, mi, km on SHIFT)**
8. LIVE **Mixed-unit arithmetic (5 ft + 7 in)**
9. LIVE **Feet-inch-fraction display with 1/2 … 1/64 precision**
10. LIVE **Fraction precision cycle 1/2 → 1/64**
11. LIVE **Convert Ans to any of 15 length / area / volume / mass units**
12. LIVE **Square units: area from length × length**
13. LIVE **Diagonal / hypotenuse Hyp(a,b)**
14. LIVE **Roof pitch Pitch(rise,run) in degrees with an x/12 ratio**
15. LIVE **Stair calculator Stair(total rise, max riser): number of risers and exact riser height**
16. LIVE **Board feet BF(thickness in, width in, length ft)**
17. LIVE **Circumference Circ(d) and circle area ACirc(d)**
18. LIVE **°C → °F and °F → °C conversions**

19. LIVE **x^2, $\sqrt{}$, x^{-1}, powers and percent**
20. LIVE **Memory M+, M−, MR and MC**
21. LIVE **One-line display with the expression echoed above the answer**
22. LIVE **Ans chaining**
23. LIVE **Bracket entry with automatic closing**
24. LIVE **History recall ▲**
25. LIVE **Cursor editing ◀ ▶ and ⌫**
26. LIVE **Fix / Sci / Eng number formats in SETUP**
27. LIVE **Fractions mixed with units (1␣2 ft + 3 in)**
28. LIVE **π and n! keys**
29. LIVE **Clear C and ON reset**
30. LIVE **Divide-by-zero and unit-mismatch errors**
31. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
32. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
33. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**
34. APP/UI **Show / hide key numbers with one button (remembered between sessions)**
35. APP/UI **Copy the current display text to the clipboard**
36. APP/UI **Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
37. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
38. APP/UI **Reference photo viewer and tool-tip name on every key**

39. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
40. PLANNED **Exam mode (locks memory & shows indicator)**
41. PLANNED **Answer history export to text**
42. PLANNED **Speech read-out of results**
43. PLANNED **Colour-blind safe key theme**
44. PLANNED **Custom shortcut keys**

2b · Feature guide

Calculator #28 is a one-line algebraic: exact fractions with a␣b␣c and measurement units (type, then =). Each card below is one of its 30 working features, with the keys that reach it and, where one exists, a verified example.

F1 Exact fraction arithmetic (no rounding): 1␣3 + 1␣6 = 1␣2

Where: #8 FT-IN Feet-inch-fraction display on/off; #13 a␣b Fraction separator (a␣b or a␣b␣c); #14 F ↔ D Fraction / mixed / decimal display

See it in action: Fraction times whole number — *3␣4 × 8* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Mixed numbers a␣b␣c entry

Where: #14 F ↔ D Fraction / mixed / decimal display

See it in action: Mixed numbers — *2␣1␣2 + 1␣1␣2* = 4

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Result cycle: improper fraction → mixed number → decimal (F ↔ D)

Where: #14 F ↔ D Fraction / mixed / decimal display; #44 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored; #8 FT-IN Feet-inch-fraction display on/off

See it in action: Add two fractions exactly (decimal view) — *1␣4 + 1␣4, then F ↔ D twice* = 0.5; Fraction times whole number — *3␣4 × 8* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Automatic lowest terms

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 A decimal point switches that calculation to decimals automatically

Where: #44 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored; #14 F ↔ D Fraction / mixed / decimal display; #32 C AC — clear all of the current calculation

See it in action: Add two fractions exactly (decimal view) — *1␣4 + 1␣4, then F ↔ D twice* = 0.5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Fraction-facts screen: mixed, decimal, numerator, denominator, percent, reciprocal

Where: #14 F ↔ D Fraction / mixed / decimal display; #8 FT-IN Feet-inch-fraction display on/off; #13 a␣b Fraction separator (a␣b or a␣b␣c)

See it in action: Add two fractions exactly (decimal view) — *1␣4 + 1␣4, then F ↔ D twice* = 0.5; Fraction times whole number — *3␣4 × 8* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Length units on keys: ft, in, m, cm, mm (yd, mi, km on SHIFT)

Where: #7 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Mixed-unit arithmetic (5 ft + 7 in)

Where: #1 ft Unit: feet; #2 in Unit: inches; #3 m Unit: metres

See it in action: Mixed numbers — *2␣1␣2 + 1␣1␣2* = 4

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Feet-inch-fraction display with 1/2 ... 1/64 precision

Where: #8 FT-IN Feet-inch-fraction display on/off; #14 F ↔ D Fraction / mixed / decimal display; #1 ft Unit: feet

See it in action: Fraction times whole number — *3␣4 × 8* = 6; Board feet — *BF(2, 6, 10)* = 10

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Fraction precision cycle 1/2 → 1/64

Where: #8 FT-IN Feet-inch-fraction display on/off; #13 a␣b Fraction separator (a␣b or a␣b␣c); #14 F ↔ D Fraction / mixed / decimal display

See it in action: Fraction times whole number — *3␣4 × 8* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Convert Ans to any of 15 length / area / volume / mass units

Where: #6 CONV Convert Ans to another unit; #11 Circ Circumference from diameter

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Square units: area from length × length

Where: #11 Circ Circumference from diameter; #15 x^2 Square x^2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Diagonal / hypotenuse Hyp(a,b)

Where: #9 Hyp Diagonal $\sqrt{(a^2+b^2)}$

See it in action: Diagonal of a rectangle — *Hyp(6, 8)* = 10

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Roof pitch Pitch(rise,run) in degrees with an x/12 ratio

Where: #9 Hyp Diagonal $\sqrt{(a^2+b^2)}$

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Stair calculator Stair(total rise, max riser): number of risers and exact riser height

Where: #10 Stair Stair risers; #21 % Percent %; #22 SETUP SETUP — angle unit, number format, display options

See it in action: Fraction times whole number — *3␣4 × 8* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Board feet BF(thickness in, width in, length ft)

Where: #10 Stair Stair risers; #1 ft Unit: feet; #8 FT-IN Feet-inch-fraction display on/off

See it in action: Board feet — *BF(2, 6, 10)*= 10

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Circumference Circ(d) and circle area ACirc(d)

Where: #11 Circ Circumference from diameter

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 °C → °F and °F → °C conversions

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 x^2, $\sqrt{}$, x^{-1}, powers and percent

Where: #21 % Percent %

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Memory M+, M−, MR and MC

Where: #23 MR MR — memory recall; #24 M− M− — subtract the value from memory M; #25 M+ M+ — add the value to independent memory M

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 One-line display with the expression echoed above the answer

Where: #8 FT-IN Feet-inch-fraction display on/off; #14 F ↔ D Fraction / mixed / decimal display; #19) Closing bracket). Closing brackets at the end of a line may be omitted

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Ans chaining

Where: #6 CONV **Convert Ans to another unit**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Bracket entry with automatic closing

Where: #19) **Closing bracket). Closing brackets at the end of a line may be omitted;** #18 (**Opening bracket (**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 History recall ▲

Where: #23 MR **MR — memory recall;** #42 ▲ **Cursor ▲ — up / previous history line / previous item**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Cursor editing ◀ ▶ and ⌫

Where: #26 ◀ Cursor ◀ — move left in the line; #27 ▶ Cursor ▶ — move right in the line; #37 ⌫ DEL — delete the character at the cursor

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Fix / Sci / Eng number formats in SETUP

Where: #22 SETUP SETUP — angle unit, number format, display options; #44 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored; #45 ± Negative sign (−) — makes the next number negative. Different from the subtraction key

See it in action: Fraction times whole number — *3⌟4 × 8* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Fractions mixed with units (1␣2 ft + 3 in)

Where: #14 F ↔ D Fraction / mixed / decimal display

See it in action: Add two fractions exactly (decimal view) — *1␣4 + 1␣4, then F ↔ D twice* = 0.5; Mixed numbers — *2␣1␣2 + 1␣1␣2* = 4

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 π and n! keys

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Clear C and ON reset

Where: #23 MR MR — memory recall; #32 C AC — clear all of the current calculation

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Divide-by-zero and unit-mismatch errors

Where: #1 ft Unit: feet; #2 in Unit: inches; #3 m Unit: metres

Try it: Press Reset, use the key(s) above, and write the display here:

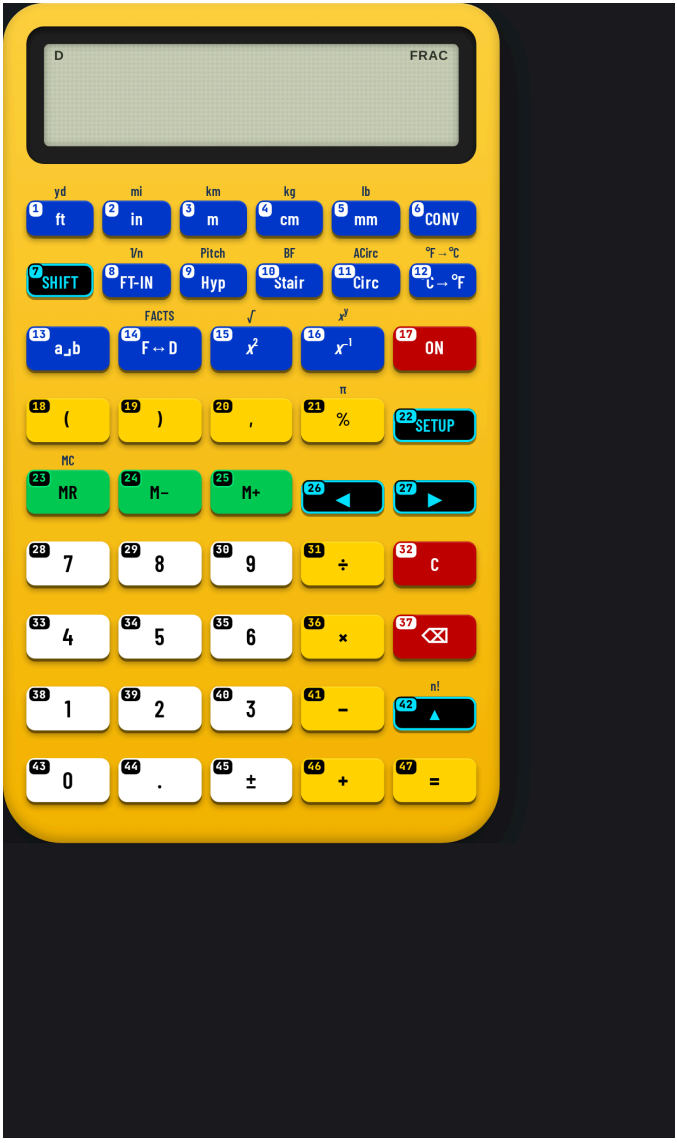
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #28 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	ft	yd	·	·	fn
2	in	mi	·	·	fn
3	m	km	·	·	fn
4	cm	kg	·	·	fn
5	mm	lb	·	·	fn
6	CONV	·	·	·	fn
7	SHIFT	·	·	·	mod
8	FT-IN	1/n	·	·	fn
9	Hyp	Pitch	·	·	fn
10	Stair	BF	·	·	fn
11	Circ	ACirc	·	·	fn
12	°C → °F	°F → °C	·	·	fn
13	a␣b	·	·	·	fn
14	F ↔ D	FACTS	·	·	fn
15	x^2	√	·	·	fn
16	x^{-1}	xy	·	·	fn
17	ON	·	·	·	clr
18	(	·	·	·	op
19	)	·	·	·	op
20	,	·	·	·	op
21	%	π	·	·	op
22	SETUP	·	·	·	mod
23	MR	MC	·	·	mem

#	Face	Shift	Alpha	3rd	Group
24	M−	·	·	·	mem
25	M+	·	·	·	mem
26	◀	·	·	·	mod
27	▶	·	·	·	mod
28	7	·	·	·	num
29	8	·	·	·	num
30	9	·	·	·	num
31	÷	·	·	·	op
32	C	·	·	·	clr
33	4	·	·	·	num
34	5	·	·	·	num
35	6	·	·	·	num
36	×	·	·	·	op
37	⌫	·	·	·	clr
38	1	·	·	·	num
39	2	·	·	·	num
40	3	·	·	·	num
41	−	·	·	·	op
42	▲	n!	·	·	mod
43	0	·	·	·	num
44	.	·	·	·	num
45	±	·	·	·	num
46	+	·	·	·	op
47	=	·	·	·	op

4 · Key-by-key reference

Every key on calculator #28, in number order, with every function it carries in this app.

#1 ft Functions

Primary: Unit: feet

shifted (2nd / f / SHIFT layer) (press #7 first): yd — Unit: yards

#2 in Functions

Primary: Unit: inches

shifted (2nd / f / SHIFT layer) (press #7 first): mi — Unit: miles

#3 m Functions

Primary: Unit: metres

shifted (2nd / f / SHIFT layer) (press #7 first): km — Unit: kilometres

#4 cm Functions

Primary: Unit: centimetres

shifted (2nd / f / SHIFT layer) (press #7 first): kg — Unit: kilograms

#5 mm Functions

Primary: Unit: millimetres

shifted (2nd / f / SHIFT layer) (press #7 first): lb — Unit: pounds

#6 CONV Functions

Primary: Convert Ans to another unit

#7 SHIFT Shift, modes, menus & navigation

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#8 FT-IN Functions

Primary: Feet-inch-fraction display on/off

shifted (2nd / f / SHIFT layer) (press #7 first): 1/n — Fraction precision 1/2 ... 1/64

#9 Hyp Functions

Primary: Diagonal $\sqrt{(a^2+b^2)}$

shifted (2nd / f / SHIFT layer) (press #7 first): Pitch — Roof pitch angle

#10 Stair Functions

Primary: Stair risers

shifted (2nd / f / SHIFT layer) (press #7 first): BF — Board feet

#11 Circ Functions

Primary: Circumference from diameter

shifted (2nd / f / SHIFT layer) (press #7 first): ACirc — Circle area from diameter

#12 °C → °F Functions

Primary: Celsius to Fahrenheit

shifted (2nd / f / SHIFT layer) (press #7 first): °F → °C — Fahrenheit to Celsius

#13 a⌟b Functions

Primary: Fraction separator (a⌟b or a⌟b⌟c)

#14 F ↔ D Functions

Primary: Fraction / mixed / decimal display

shifted (2nd / f / SHIFT layer) (press #7 first): FACTS — Fraction facts

#15 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #7 first): √ — Square root √

Keyboard: q

#16 x^{-1} Functions

Primary: Reciprocal x^{-1} (1/x)

shifted (2nd / f / SHIFT layer) (press #7 first): xy — Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: i

#17 ON `Clear, delete & power`

Primary: ON — power on

#18 (`Operators & execute`

Primary: Opening bracket (

Keyboard: (

#19) `Operators & execute`

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#20 , `Operators & execute`

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#21 % `Operators & execute`

Primary: Percent %

shifted (2nd / f / SHIFT layer) (press #7 first): π — π (3.14159265…) as an exact constant

Keyboard: %

#22 SETUP `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

#23 MR `Memory, variables & registers`

Primary: MR — memory recall

shifted (2nd / f / SHIFT layer) (press #7 first): MC — Clear memory

#24 M− `Memory, variables & registers`

Primary: M− — subtract the value from memory M

#25 M+ `Memory, variables & registers`

Primary: M+ — add the value to independent memory M

#26 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

Keyboard: `ArrowLeft`

#27 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

Keyboard: `ArrowRight`

#28 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#29 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#30 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#31 ÷ Operators & execute

Primary: Division ÷

Keyboard: /

#32 C Clear, delete & power

Primary: AC — clear all of the current calculation

Keyboard: Escape

#33 4 Digits & number entry

Primary: Digit 4

Keyboard: 4

#34 **5** `Digits & number entry`

Primary: Digit 5

Keyboard: `5`

#35 **6** `Digits & number entry`

Primary: Digit 6

Keyboard: `6`

#36 **×** `Operators & execute`

Primary: Multiplication ×

Keyboard: `*` `x`

#37 ⌫ `Clear, delete & power`

Primary: DEL — delete the character at the cursor

Keyboard: `Backspace` `Delete`

#38 **1** `Digits & number entry`

Primary: Digit 1

Keyboard: `1`

#39 **2** `Digits & number entry`

Primary: Digit 2

Keyboard: `2`

#40 **3** Digits & number entry

Primary: Digit 3

Keyboard: 3

#41 **–** Operators & execute

Primary: Subtraction –

Keyboard: -

#42 **▲** Shift, modes, menus & navigation

Primary: Cursor ▲ — up / previous history line / previous item

shifted (2nd / f / SHIFT layer) (press #7 first): n! — Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: ArrowUp

#43 **0** Digits & number entry

Primary: Digit 0

Keyboard: 0

#44 **.** Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: .

#45 **±** `Digits & number entry`

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

#46 **+** `Operators & execute`

Primary: Addition +

Keyboard: `+`

#47 **=** `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

5 · Worked examples

6 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Add two fractions exactly (decimal view)

Calculate **1␣4 + 1␣4**, then **F ↔ D** twice. Press **Reset** first.

#38 1 → #13 a␣b → #33 4 → #46 + → #38 1 → #13 a␣b → #33 4 → #47 = → #14 F ↔ D → #14 F ↔ D

Display shows `0.5` — expected 0.5. ✔ verified on the app.

Example 5.2 — Fraction times whole number

Calculate **3␣4 × 8**. Press **Reset** first.

#40 3 → #13 a␣b → #33 4 → #36 × → #29 8 → #47 =

Display shows `6` — expected 6. ✔ verified on the app.

Example 5.3 — Diagonal of a rectangle

Calculate **Hyp(6, 8)**. Press **Reset** first.

#9 Hyp → #35 6 → #20 , → #29 8 → #19) → #47 =

Display shows `10` — expected 10. ✔ verified on the app.

Example 5.4 — Board feet

Calculate **BF(2, 6, 10)**. Press **Reset** first.

#7 SHIFT → #10 BF → #39 2 → #20 , → #35 6 → #20 , → #38 1 → #43 0 → #19) → #47 =

Display shows `10` — expected 10. ✔ verified on the app.

Example 5.5 — Celsius to Fahrenheit

Calculate **100 °C → °F**. Press **Reset** first.

#12 °C→°F → #38 1 → #43 0 → #43 0 → #19) → #47 =

Display shows `212` — expected 212. ✔ verified on the app.

Example 5.6 — Mixed numbers

Calculate **2␣1␣2 + 1␣1␣2**. Press **Reset** first.

#39 2 → #13 a␣b → #38 1 → #13 a␣b → #39 2 → #46 + → #38 1 → #13 a␣b → #38 1 → #13 a␣b → #39 2 → #47 =

Display shows `4` — expected 4. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

16 extra functions sit on shifted layers. Press the layer key once, then the key: for example #7 then #1 → yd; #7 then #2 → mi; #7 then #3 → km; #7 then #4 → kg.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #47.

Hack 13: Type faster on a computer

27 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
q	#15	x^2	Square x^2
i	#16	x^{-1}	Reciprocal x^{-1} (1/x)
(	#18	(	Opening bracket (
)	#19	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#20	,	Comma / argument separator (between function arguments, list items or coordinates)
%	#21	%	Percent %
ArrowLeft	#26	◀	Cursor ◀ — move left in the line
ArrowRight	#27	▶	Cursor ▶ — move right in the line
7	#28	7	Digit 7
8	#29	8	Digit 8
9	#30	9	Digit 9
/	#31	÷	Division ÷
Escape	#32	C	AC — clear all of the current calculation
4	#33	4	Digit 4
5	#34	5	Digit 5
6	#35	6	Digit 6
* x	#36	×	Multiplication ×
Backspace Delete	#37	⌫	DEL — delete the character at the cursor
1	#38	1	Digit 1
2	#39	2	Digit 2

3	#40	3	Digit 3
-	#41	–	Subtraction –
ArrowUp	#42	▲	Cursor ▲ — up / previous history line / previous item
0	#43	0	Digit 0
.	#44	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
+	#46	+	Addition +
Enter =	#47	=	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 9: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (6)

Work each one on calculator #28, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 1␣4 + 1␣4, then F ↔ D twice = _____
2. 3␣4 × 8 = _____
3. Hyp(6, 8) = _____
4. BF(2, 6, 10) = _____
5. 100 °C → °F = _____
6. 2␣1␣2 + 1␣1␣2 = _____

Set B — Key-finding drills (53)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “ft” (Unit: feet)? _____
2. Which key is “in” (Unit: inches)? _____
3. Which key is “m” (Unit: metres)? _____
4. Which key is “cm” (Unit: centimetres)? _____
5. Which key is “mm” (Unit: millimetres)? _____
6. Which key is “CONV” (Convert Ans to another unit)? _____
7. Which key is “SHIFT” (SHIFT / 2nd)? _____
8. Which key is “FT-IN” (Feet-inch-fraction display on/off)? _____

9. **Which key is “Hyp” (Diagonal √)?** _____
10. **Which key is “Stair” (Stair risers)?** _____
11. **Which key is “Circ” (Circumference from diameter)?** _____
12. **Which key is “°C → °F” (Celsius to Fahrenheit)?** _____
13. **Which key is “a␣b” (Fraction separator)?** _____
14. **Which key is “F ↔ D” (Fraction / mixed / decimal display)?** _____
15. **Which key is “x^2” (Square x^2)?** _____
16. **Which key is “x^{-1}” (Reciprocal x^{-1})?** _____
17. **Which key is “ON” (ON)?** _____
18. **Which key is “(” (Opening bracket)?** _____
19. **Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)?** _____
20. **Which key is “,” (Comma / argument separator)?** _____
21. **Which key is “%” (Percent %)?** _____
22. **Which key is “SETUP” (SETUP)?** _____
23. **Which key is “MR” (MR)?** _____
24. **Which key is “M−” (M−)?** _____
25. **Which key is “M+” (M+)?** _____
26. **Which key is “◀” (Cursor ◀)?** _____
27. **Which key is “▶” (Cursor ▶)?** _____
28. **Which key is “÷” (Division ÷)?** _____

29. Which key is “C” (AC)? _____

30. Which key is “×” (Multiplication ×)? _____

31. Which key is “⌫” (DEL)? _____

32. Which key is “−” (Subtraction −)? _____

33. Which key is “ ▲ ” (Cursor ▲)? _____

34. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

35. Which key is “±” (Negative sign)? _____

36. Which key is “+” (Addition +)? _____

37. Which key is “=” (Execute / =)? _____

38. How do you reach “yd”? _____

39. How do you reach “mi”? _____

40. How do you reach “km”? _____

41. How do you reach “kg”? _____

42. How do you reach “lb”? _____

43. How do you reach “1/n”? _____

44. How do you reach “Pitch”? _____

45. How do you reach “BF”? _____

46. How do you reach “ACirc”? _____

47. How do you reach “°F → °C”? _____

48. How do you reach “FACTS”? _____

49. How do you reach "√"? _____

50. How do you reach "xy"? _____

51. How do you reach "π"? _____

52. How do you reach "MC"? _____

53. How do you reach "n!"? _____

Appendix A — Drill answers

Set A

1. **0.5** (display `0.5`)

   `#38` 1 → `#13` a␣b → `#33` 4 → `#46` + → `#38` 1 → `#13` a␣b → `#33` 4 → `#47` = → `#14` F↔D → `#14` F↔D

2. **6** (display `6`)

   `#40` 3 → `#13` a␣b → `#33` 4 → `#36` × → `#29` 8 → `#47` =

3. **10** (display `10`)

   `#9` Hyp → `#35` 6 → `#20` , → `#29` 8 → `#19`) → `#47` =

4. **10** (display `10`)

   `#7` SHIFT → `#10` BF → `#39` 2 → `#20` , → `#35` 6 → `#20` , → `#38` 1 → `#43` 0 → `#19`) → `#47` =

5. **212** (display `212`)

   `#12` °C→°F → `#38` 1 → `#43` 0 → `#43` 0 → `#19`) → `#47` =

6. **4** (display `4`)

   `#39` 2 → `#13` a␣b → `#38` 1 → `#13` a␣b → `#39` 2 → `#46` + → `#38` 1 → `#13` a␣b → `#38` 1 → `#13` a␣b → `#39` 2 → `#47` =

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #31
29. #32
30. #36
31. #37
32. #41
33. #42
34. #44
35. #45
36. #46
37. #47
38. #7 SHIFT then #1 ft
39. #7 SHIFT then #2 in
40. #7 SHIFT then #3 m
41. #7 SHIFT then #4 cm
42. #7 SHIFT then #5 mm
43. #7 SHIFT then #8 FT-IN
44. #7 SHIFT then #9 Hyp
45. #7 SHIFT then #10 Stair
46. #7 SHIFT then #11 Circ
47. #7 SHIFT then #12 °C→°F
48. #7 SHIFT then #14 F↔D
49. #7 SHIFT then #15 x^2
50. #7 SHIFT then #16 x^{-1}
51. #7 SHIFT then #21 %
52. #7 SHIFT then #23 MR
53. #7 SHIFT then #42 ▲

Appendix B — Function finder (A–Z)

Function	Label	Keys
AC — clear all of the current calculation	C	#32
Addition +	+	#46
Board feet	BF	#7 → #10
Celsius to Fahrenheit	°C → °F	#12
Circle area from diameter	ACirc	#7 → #11
Circumference from diameter	Circ	#11
Clear memory	MC	#7 → #23
Closing bracket). Closing brackets at the end of a line may be omitted	)	#19
Comma / argument separator (between function arguments, list items or coordinates)	,	#20
Convert Ans to another unit	CONV	#6
Cursor ▲ — up / previous history line / previous item	▲	#42
Cursor ▶ — move right in the line	▶	#27
Cursor ◀ — move left in the line	◀	#26
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#44
DEL — delete the character at the cursor	⌫	#37
Diagonal $\sqrt{(a^2+b^2)}$	Hyp	#9
Digit 0	0	#43
Digit 1	1	#38
Digit 2	2	#39
Digit 3	3	#40
Digit 4	4	#33
Digit 5	5	#34

Digit 6	6	#35
Digit 7	7	#28
Digit 8	8	#29
Digit 9	9	#30
Division ÷	÷	#31
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	=	#47
Factorial n! (also on non-integers via the gamma function where supported)	n!	#7 → #42
Fahrenheit to Celsius	°F → °C	#7 → #12
Feet-inch-fraction display on/off	FT-IN	#8
Fraction / mixed / decimal display	F ↔ D	#14
Fraction facts	FACTS	#7 → #14
Fraction precision 1/2 ... 1/64	1/n	#7 → #8
Fraction separator (a␣b or a␣b␣c)	a␣b	#13
M+ — add the value to independent memory M	M+	#25
M− — subtract the value from memory M	M−	#24
MR — memory recall	MR	#23
Multiplication ×	×	#36
Negative sign (−) — makes the next number negative. Different from the subtraction key	±	#45
ON — power on	ON	#17
Opening bracket (	(	#18
Percent %	%	#21
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xy	#7 → #16
Reciprocal x^{-1} (1/x)	x^{-1}	#16
Roof pitch angle	Pitch	#7 → #9
SETUP — angle unit, number format, display options	SETUP	#22

SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#7
Square root √	√	#7 → #15
Square x^2	x^2	#15
Stair risers	Stair	#10
Subtraction –	–	#41
Unit: centimetres	cm	#4
Unit: feet	ft	#1
Unit: inches	in	#2
Unit: kilograms	kg	#7 → #4
Unit: kilometres	km	#7 → #3
Unit: metres	m	#3
Unit: miles	mi	#7 → #2
Unit: millimetres	mm	#5
Unit: pounds	lb	#7 → #5
Unit: yards	yd	#7 → #1
π (3.14159265…) as an exact constant	π	#7 → #21

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

One-line fraction & measurement calculator for workshop, building and kitchen work. Whole-number and fraction input stays exact; any decimal point switches that calculation to decimals.

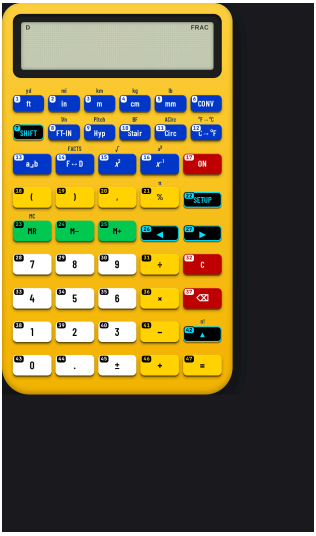
Units use the bundled math.js unit system; the feet-inch-fraction display, stair, pitch and board-feet functions are U-45 code.

Planned (not yet working)

- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 ft ⇧ yd	#2 in ⇧ mi	#3 m ⇧ km
#4 cm ⇧ kg	#5 mm ⇧ lb	#6 CONV
#7 SHIFT	#8 FT-IN ⇧ 1/n	#9 Hyp ⇧ Pitch
#10 Stair ⇧ BF	#11 Circ ⇧ ACirc	#12 °C → °F ⇧ °F → °C
#13 a␣b	#14 F ↔ D ⇧ FACTS	#15 x^2 ⇧ √
#16 x^{-1} ⇧ xy	#17 ON	#18 (
#19)	#20 ,	#21 % ⇧ π
#22 SETUP	#23 MR ⇧ MC	#24 M−
#25 M+	#26 ◀	#27 ▶
#28 7	#29 8	#30 9
#31 ÷	#32 C	#33 4
#34 5	#35 6	#36 ×
#37 ⌫	#38 1	#39 2
#40 3	#41 −	#42 ▲ ⇧ n!
#43 0	#44 .	#45 ±
#46 +	#47 =	

U-45 CALCULATORS · Calculator #28 · cut out and keep